

Lingam Hemang

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Professional Summary

GenAI Engineer with **4 years** at Infosys shipping **production AI applications and data systems** for **Apple Inc.** Built and deployed end-to-end **RAG pipelines, LLM-powered APIs, and agentic AI tools** using LangChain, AWS Bedrock, and ChromaDB. Backed by deep expertise in **distributed data engineering** (Spark, Kafka, Airflow, Snowflake) that enables building AI systems grounded in reliable, production-grade data infrastructure.

Technical Skills

GenAI & LLM: AWS Bedrock, LangChain, RAG Pipelines, Vector Embeddings, ChromaDB, Ollama, Prompt Engineering, LLM Application Development, Agentic AI, SSE Streaming

Languages: Python, SQL, PySpark

Backend & APIs: Django, Flask, REST API Design & Integration, Server-Sent Events (SSE), OOP, SOLID Principles, Design Patterns, pytest

Data Engineering: Apache Spark, Apache Kafka, Apache Airflow, Snowflake, AWS S3, AWS Lambda, ETL Pipelines, Data Modelling, Schema Design & Validation, Partitioning & Query Optimisation

ML: XGBoost, LightGBM, scikit-learn, Plotly, Dash

DevOps & Tools: Docker, Git, CI/CD, Linux, Bash, AWS Lambda, Splunk, Agile/Scrum, Jira, Confluence

Projects

Agentic AI Git Diff Analyzer | *Python, Django, AWS Bedrock, AWS Lambda, Ollama, ReportLab* | [Repo](#)

- **Dual-mode agentic SQL schema impact analyzer** – local (Ollama) and **cloud mode** (AWS Bedrock Claude 3.5 via GitHub Webhook → Lambda → SSM → EC2); **LLM-generated impact reports** with risk-level badges (CRITICAL → LOW) as styled A4 PDFs; commit-accurate downstream dependency detection; real-time **Django UI with SSE** streaming and run deduplication by commit hash.

Document Intelligence Chatbot – RAG Pipeline | *Python, Django, LangChain, ChromaDB, Ollama, EasyOCR* | [Repo](#)

- End-to-end **RAG pipeline** (LangChain + ChromaDB + Ollama) with **persistent multi-turn conversation memory**; multi-format document ingestion (PDF, DOCX, TXT, Images) via PyMuPDF + EasyOCR with auto-rotation; **incremental vector store** with timestamped snapshots and rollback; **10-endpoint Django REST API** supporting live model switching and VectorDB management.

Retail ML Sales Forecasting API | *Python, Flask, scikit-learn, XGBoost, LightGBM* | [Repo](#) **Infosys Hackathon – 2nd Place**

- Multi-model **ML forecasting system** (Random Forest, XGBoost, LightGBM) for sales, store projections, and product recommendations via **Flask REST API**; cyclical sin/cos month encoding for seasonal patterns; cross-join **purchase probability scoring**; interactive **Dash + Plotly** dashboard with live reactive callbacks.

Experience

Infosys

Hyderabad, India

Specialist Programmer – GenAI Engineer

July 2022 – Present

Client: Apple Inc. | **Python, Spark, Kafka, Airflow, Snowflake, AWS S3, AWS Lambda, PySpark, Splunk**

- Engineered production **ETL pipelines** on a large-scale **distributed data platform** (Spark, Kafka, Airflow, AWS); built **Python data services** consuming internal and external **REST APIs** to ingest data into **Snowflake**; query optimisation cut end-to-end processing time by **40%**.
- Designed **high-availability, fault-tolerant ingestion pipelines** across **AWS S3** and **Kafka event streams** via **Airflow DAGs** with retry logic and SLA monitoring – achieving **99.5% uptime**; schema validation and self-healing scripts reduced pipeline error rate by **35%**.
- Deployed a **client ML name-matching model** to the Airflow cluster – provisioning input views, configuring connections, and managing execution; built **semantic layer processes** with materialised views for downstream analytics and **Tableau** dashboards.
- Provided **backend data support** for a production application – maintained **database view consistency** and loaded data into a **semantic search cluster**; managed releases via **CI/CD** pipelines; developed reusable **framework extensions** cutting data source onboarding by **30%**.

Education

SRM Institute of Science and Technology

May 2018 – June 2022

Bachelor of Technology – Electronics and Communication Engineering

CGPA: 9.52/10.0